Unit

3

Lesson

1

Beyond Imagination.

Into the Emergency Simulator.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Explain the basic concept of short-range Bluetooth communication for data transfer.
- Connect and correctly wire a Bluetooth module (e.g., HC-05/06) to an Arduino.
- Design a basic App Inventor interface to connect to a Bluetooth device.

Challenge questions of the lesson.

- How does short-range Bluetooth communication transfer data between devices, and what are its range limitations?
- What are the correct steps to wire a Bluetooth module (e.g., HC-05/06) to an Arduino for communication?
- How can you design a user interface in MIT App Inventor to pair and communicate with a Bluetooth device?
- What are some common issues you might encounter when connecting an Arduino to a Bluetooth module, and how can you troubleshoot them?

SEL Relation

- **Social Awareness:** Cultivating profound empathy by designing technology-driven solutions that address critical community safety needs and cater to the diverse requirements of all individuals.
- **Responsible Decision-Making:** Grappling with the ethical implications, reliability considerations, and potential biases inherent in creating autonomous smart alert systems, especially for critical emergency responses.
- **Relationship Skills:** Engaging in complex collaborative design and problem-solving, practicing constructive critique and effective communication within a team setting.
- **Self-Management:** Demonstrating resilience and perseverance in tackling highly complex coding and system integration challenges, particularly when implementing pattern recognition algorithms.
- **Self-Awareness:** Reflecting on their evolving capabilities as engineers and designers in addressing significant societal challenges through technology.

Engage

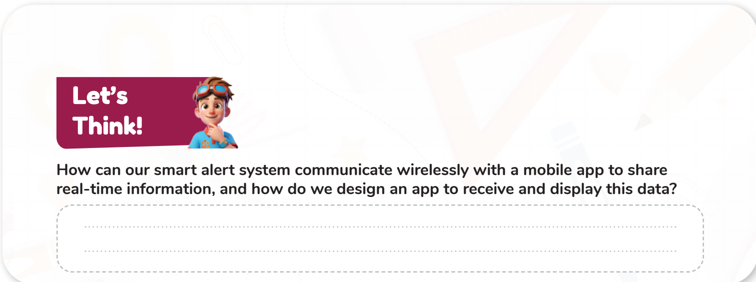
Teacher's Role:

- Present the core question clearly to spark curiosity and relevance.
- Facilitate a brainstorming session to map out wireless communication methods, data flow, app design elements, and benefits.
- Guide students in understanding the differences between Bluetooth, Wi-Fi, and other wireless options in the context of alert systems.
- Support students in considering user experience and accessibility for diverse users.
- Encourage real-world connections by discussing existing IoT devices and their mobile apps.

Expected from students:

- Actively participate in the brainstorming session by contributing ideas and questions.
- Demonstrate understanding of wireless communication basics by discussing Bluetooth, Wi-Fi, and alternatives.
- Analyze data flow from sensors to mobile apps and identify critical design elements.
- Consider accessibility and usability when imagining mobile app features.
- Share insights during discussions and collaborate with peers during activities.

Parts of the book included.


Let's Think!

How can our smart alert system communicate wirelessly with a mobile app to share real-time information, and how do we design an app to receive and display this data?

Let's Think!

Objective:

Students will explore how a smart alert system can communicate wirelessly with a mobile app to share real-time sensor data and learn key design considerations for building an effective and accessible app interface.

Answers:

Wireless Communication Methods:

- **Bluetooth (BLE):** Ideal for short-range communication directly with a phone; low power consumption.
- **Wi-Fi (ESP32/ESP8266):** Connects to local networks and the internet, enabling cloud integration and broader reach.
- **Other Technologies:** Briefly mention LoRa (long-range) and Cellular (wide-area) communications, noting their complexity.

Data Flow:

- Sensors collect data (e.g., gas levels, temperature).
- Arduino processes data and determines when to send alerts.
- Wireless module transmits data.
- Mobile app receives and interprets the data.

Mobile App Design Considerations:

- **Purpose:** What information does the user need? (e.g., current readings, alert status, historical data)
- **User Interface (UI):** How will data be displayed? (numbers, graphs, gauges, colors) How will alerts look and sound?
- **User Experience (UX):** Ease of understanding and use, especially in emergencies. Accessibility for different abilities (large text, clear icons, voice alerts).
- **Development Tools:** Visual programming like App Inventor or Thunkable, or advanced coding options.
- **Notifications:** How will the app push alerts even when closed?

Benefits of App Integration:

- Remote monitoring without physical proximity.
- Personalized alerts with different sounds or vibrations.
- Data logging and analysis over time.
- Accessibility for various user needs (e.g., visual or auditory impairments).
- Broader alert distribution to multiple users (family, caregivers).

Possible Strategies for Implementation:

1. Wireless Connections Jigsaw Research:

- Divide students into small groups, each researching a wireless technology (Bluetooth, Wi-Fi, etc.).
- Groups summarize key principles, advantages, disadvantages, and use cases.
- Groups present findings to the class to facilitate peer learning.

2. App Sketching Activity:

- Provide blank phone screen templates or plain paper.
- Students sketch mobile app designs for the multi-sensory alert system, including data displays and alerts.

- Encourage multiple screen sketches (dashboard, settings, alert history).
- Students share and discuss their designs.

3. **Think-Pair-Share: "Why Go Wireless?"**

- Pose the question: "Why is wireless communication essential for modern smart alert systems, and who benefits most?"
- Students think individually, discuss in pairs, then share insights with the class.
- Supports critical thinking and articulation skills.

4. **Case Study Discussion:**

- Present real-world IoT examples (e.g., smart smoke detectors, fitness trackers).
- Discuss wireless communication methods and app data displays.
- Connect theoretical concepts to practical applications.

Explorer's Toolkit

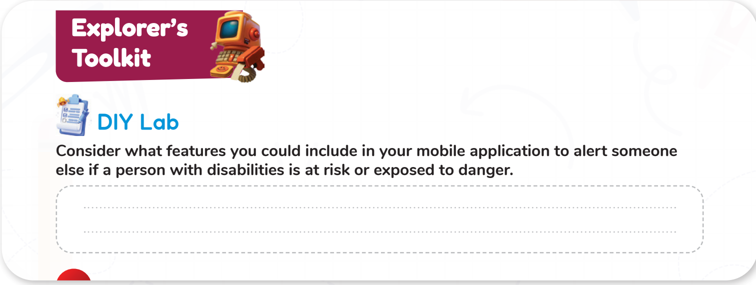
Teacher's Role:

- Introduce the scenario by posing a real-world design challenge:
- "What if someone with a disability is alone and needs urgent help? How can technology bridge that gap and notify someone else in time?"
- Frame this as a problem-solving and empathy challenge, not just a creative one. Emphasize that good design considers limitations, risks, and accessibility.
- Encourage students to think in terms of inputs, sensors, triggers, and user interaction.
- Guide discussion toward feasibility and impact, prompting students to evaluate their ideas based on the needs of real users.

Expected from Students:

- Analyze potential risk scenarios for people with disabilities (e.g., falls, disorientation, getting stuck, medical emergencies).
- Propose realistic app features that use sensors or user input to send alerts or initiate emergency responses.
- Justify their feature ideas based on real user needs and potential safety benefits.
- Document their ideas clearly in writing, using specific terms like "Bluetooth-triggered signal," "voice-activated command," or "accelerometer-based fall detection."
- Prepare to explain how each proposed feature would function and how it addresses a specific user challenge.

Parts of the book included.


Explorer's Toolkit

DIY Lab

Consider what features you could include in your mobile application to alert someone else if a person with disabilities is at risk or exposed to danger.

DIY Lab

Objective:

Students will evaluate potential safety risks faced by individuals with disabilities and design technology-based features to alert caregivers or emergency contacts during high-risk situations. They will think critically about usability, accessibility, and the technical tools available in mobile apps to develop user-centered alert systems.

Answers:

Question:

What features could you include in your mobile application to alert someone else if a person with disabilities is at risk or exposed to danger?

Possible Responses:

1. **Voice-Triggered Help Alert**
   - If the user says "Help me," the app sends a pre-programmed SMS to emergency contacts with GPS location.
2. **Accelerometer-Based Fall Detection**
   - The phone detects a sudden fall and automatically initiates a timer. If the user doesn't cancel the timer, it sends an alert.
3. **Inactivity Timer**
   - If no activity is detected for a set time (e.g., 3 hours), the app sends a silent background message to check in.
4. **Shake-to-Alert Feature**
   - The user can shake the phone 3 times quickly to activate an emergency alert without navigating menus.
5. **Bluetooth Beacon Pairing**
   - When the user is out of Bluetooth range from a wearable device, the app assumes a separation and sends a warning.

Possible Strategies for Implementation:

1. **Design Thinking Scenario Setup**
   - Present a user case study:

     "Sara is 65, visually impaired, and lives alone. What could go wrong—and what would your app need to detect it?"
   - Use this scenario to build deeper problem-context understanding before students brainstorm.
2. **Categorize Ideas Using Systems Thinking**
   - After students brainstorm, have them categorize their ideas by:
     - Input type: voice, motion, time, manual button
     - Alert type: SMS, call, sound, push notification
     - User type: visual impairment, limited mobility, memory loss

     This helps them visualize system flow and user interaction.
3. **Peer Collaboration and Critique**
   - Encourage small groups to develop 2–3 ideas and then exchange with another group to give feedback:
     - "Is this feasible?"
     - "Would this make sense for the user?"
     - "Could this alert system be triggered accidentally?"
4. **Compare with Real Tech Solutions**
   - Show clips or demos of actual alert apps/devices (e.g., LifeAlert, Apple Fall Detection, wearable SOS apps). Ask students to:
     - Identify what works well.
     - Propose improvements or modifications.
5. **Visual Sketch Extension (Optional)**
   - Ask students to sketch wireframes or UX flows of how their feature would be used in an app.
   - Prompt them to annotate:
     - Trigger condition
     - What the app sends or does
     - Who receives the alert

Explain

Teacher's Role:

- Introduce the concept of Bluetooth as a wireless communication tool that enables devices to exchange data without cables—essential in assistive apps and alert systems.
- Prompt students to reflect on where they've encountered Bluetooth before (e.g., headphones, speakers, smartwatches).
- Explain that this video will help them understand how Bluetooth functions, especially in the context of apps they'll be building later (e.g., transmitting alert messages between devices).
- Pause the video at key moments to clarify technical terms and check for understanding.
- Facilitate a short discussion or recap after the video to help students connect what they learned to their app design goals.

Expected from Students:

- Watch the video actively and take notes on how Bluetooth enables wireless data transfer between devices.
- Identify key terms (e.g., pairing, signal, range, interference).
- Reflect on how Bluetooth could be applied in a real-world context such as an app that sends alerts or connects to wearable devices.
- Be prepared to answer questions or discuss how Bluetooth will support user safety in their app.

Parts of the book included.

Screen Lessons

Let's watch this video to learn more about the Bluetooth and how it works:

https://www.youtube.com/watch?v=JLJfRpfZ5lY

Screen Lessons

Objective:

Students will understand the core principles behind Bluetooth communication, including how devices pair, how data is exchanged, and what limitations (e.g., range, signal interference) exist. This knowledge will help students design apps that use Bluetooth to connect assistive features across devices.

Video Link: http://youtube.com/watch?v=jzxZUJmOu3o&feature=youtu.be

Possible Strategies for Implementation:

1. **Pre-Watching**

- Activate Prior Knowledge: Ask students:
  - o "Have you ever used Bluetooth? What did you use it for?"
  - o "What do you think happens when two devices connect?"
- Preview Vocabulary:
  - o Introduce and define key terms: pairing, frequency, range, signal interference, wireless communication.
- Set a Purpose for Viewing: Write this focus question on the board:

  "How does Bluetooth allow one device to send a message to another without wires?"

2. **During Watching**

- Segmented Viewing: Pause at major points in the video (e.g., explanation of signal, pairing process, or range limitations) and ask clarifying questions:
  - o "What makes Bluetooth different from Wi-Fi?"
  - o "What kind of data can be sent using Bluetooth?"
- Note-Taking Guide: Provide a structured note sheet with columns such as:
  - o Term, What It Means, Example in Real Life
- Think-Pair-Share: After key segments, ask students to turn to a partner and discuss:
  - o "How does this connect to the reminder app we're building?"
  - o "Would Bluetooth be useful for sending alerts? Why?"

3. **Post-Watching**

- Class Discussion: Facilitate reflection on how Bluetooth could be used in:
  - o A medicine alert system
  - o A fall detector paired with a caregiver's phone
- Quick Recap Questions:
  - o "What's the average range of Bluetooth?"
  - o "Why might Bluetooth fail or be unreliable?"
- Application Prompt: Ask students to answer:

  "If you were designing an alert system for someone with a disability, how would Bluetooth help make it work?"
- Exit Ticket (Optional):

  "One thing I learned about Bluetooth today is..."

  "One way I could use Bluetooth in my app is..."

Elaborate and Evaluate

Teacher's Role:

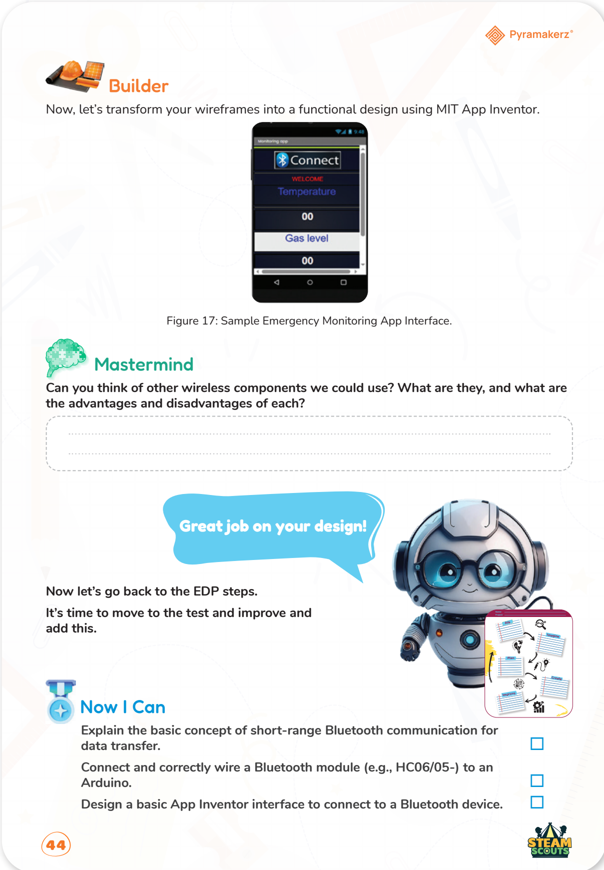
- Explain to students the importance of wireframing in app development, emphasizing how it helps visualize the app's layout and user flow.
- Introduce Excalidraw as a simple tool to create wireframes, and demonstrate its features.
- Provide an example of a wireframe for a basic app screen, such as the reminder screen for the medicine app.
- Encourage students to focus on the placement of key components like buttons, text boxes, and images. Remind them to think about how these components will interact.
- Guide students in transforming their wireframes into a functional app using MIT App Inventor.
- Show students how to map the wireframe's layout to the actual screen in MIT App Inventor, including the use of arrangements, components, and properties.
- Provide support for students as they integrate their wireframe designs and help them troubleshoot any design or coding issues.
- Introduce the discussion about different wireless components like Wi-Fi, NFC, and Bluetooth.
- Encourage students to brainstorm other wireless options that could be useful in their apps and evaluate their strengths and weaknesses.
- Lead a class discussion where students share their ideas and reasoning behind their chosen wireless component.

Expected from Students:

- Students should create a detailed wireframe using Excalidraw, showing the layout of their app interface, including:
  - Button placement (e.g., "Set Reminder")
  - Text fields (e.g., input for reminder time and message)
  - Visual components (e.g., alarm image)
- They should ensure their wireframe is clean, readable, and reflects the intended functionality of their app.
- Students will use MIT App Inventor to bring their wireframes to life by building the app layout, ensuring that all components are placed correctly and function as intended.
- They should connect their layout with the app's logic (e.g., Button.Click events, TextBox inputs).
- Test their app's user interface for usability, ensuring it is intuitive and meets their user persona's needs.

- Students should identify at least one additional wireless component that could enhance their app and provide reasoning for why they chose it.
- They should consider the advantages (e.g., range, ease of use) and disadvantages (e.g., power consumption, compatibility) of the wireless component and its potential impact on the app's functionality.
- Students will be expected to share their findings with the class and engage in the discussion by presenting their ideas and listening to others' suggestions.

Parts of the book included.


Pyramakerz

Builder

Now, let's transform your wireframes into a functional design using MIT App Inventor.

Figure 17: Sample Emergency Monitoring App Interface.

Mastermind

Can you think of other wireless components we could use? What are they, and what are the advantages and disadvantages of each?

Great job on your design!

Now let's go back to the EDP steps.

It's time to move to the test and improve and add this.

Now I Can

- Explain the basic concept of short-range Bluetooth communication for data transfer. ☐
- Connect and correctly wire a Bluetooth module (e.g., HC06/05-) to an Arduino. ☐
- Design a basic App Inventor interface to connect to a Bluetooth device. ☐

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STEAM SCOUTS

Apprentice

Objective:

Students will design and visualize the interface of their medicine reminder app by creating a wireframe. Using Excalidraw, they will map out the layout of their app, considering where each component (buttons, text fields, images) should be placed. This wireframe will serve as a blueprint for the app development process in the next phases.

Link: https://excalidraw.com/

Activity Instructions:

1. **Open Excalidraw**

- Go to Excalidraw.
- You don't need an account, but you can save your wireframe by clicking the Save

button at the top (or copy the link to your design).

- Select New Drawing to start with a blank canvas.

2. **Set the Canvas Size**

- Adjust the canvas size to match the layout you want for your app screen.
- For simplicity, make it 720px by 1280px (standard mobile screen resolution).

3. **Start with Basic Structure**

- Draw a Rectangle to represent the overall screen layout (this is the outline of your app).
- Add sections or blocks within this rectangle for the different elements of your app (e.g., button area, input area, image section).

4. **Add Key Components**

- Add Text Fields: Use the Text tool to label and show where text input fields (e.g., for entering time, alert message) will go.
- Add Buttons: Use rectangles or rounded boxes to represent buttons (e.g., “Set Reminder”).
- Add Images: Represent images (e.g., alarm icon) with smaller rectangles and label them appropriately.

5. **Label the Components**

- Add labels near each component to explain what they are. For example, near the button, write "Set Reminder" and near the input boxes, label them "Enter Time" or "Alert Message".

6. **Visualize User Flow**

- Think about the user's journey. Where will they click first? What is the next action?
- Draw arrows to show how the user would interact with the app (e.g., from the text box to the “Set Reminder” button).

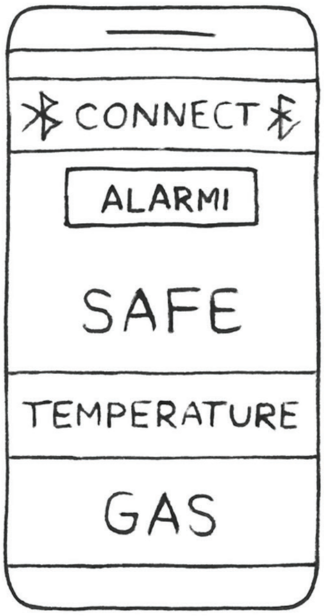
7. **Refine and Finalize Your Design**

- Once you have all your components in place, take a step back and ensure the design makes sense and is easy to use.
- Check that components are well-spaced, clearly labeled, and logically arranged.

8. **Save Your Wireframe**

- Once satisfied, click Save in Excalidraw. Share the link with your teacher or download the image for submission.

Possible sketch:



Possible Strategies for Implementation:

1. **Pre-Activity Preparation**

- Introduce the Purpose of Wireframing:
  - Explain the importance of planning the app's user interface (UI) before actual coding begins.
  - Discuss how wireframing helps visualize the app layout and plan its user flow.
  - Explain the terms wireframe, UI components, and user flow in the context of the app design.
- Provide Sample Wireframe Example:
  - Show a simple example of a wireframe to demonstrate how different components (buttons, input fields, images) should be arranged on the app screen.
  - Ensure students understand how to organize components to ensure ease of use for the end-user.

2. **Using Excalidraw**

- Walkthrough of Excalidraw Tool:
  - Provide a brief tutorial on using Excalidraw. Show students how to create shapes (rectangles for buttons, text boxes, etc.) and group components (e.g., labels, input fields).
  - Demonstrate how to use lines and arrows to show the interaction flow (e.g., from a "Connect" button to the Bluetooth functionality).
- Explain Components to Include:
  - Have students include key components such as:

    Alarm Image: Represented as an icon for the alert.

Textboxes: For input like time, gas readings, etc.

Buttons: To connect the app to the Bluetooth or activate a function.

Labels: To describe inputs or display real-time readings.

3. **Encourage Iteration and Refinement**
   - Allow for Revisions:
     - Encourage students to revise and refine their wireframe as they work. Emphasize that wireframes are not final designs but drafts that can change.
     - Remind students to focus on usability—ensure that the app's flow is simple and intuitive.
   - Peer Feedback:
     - Have students pair up and review each other's wireframes. They can provide feedback on design clarity, logical flow, and ease of use.
4. **Simplified User Experience Focus**
   - Prioritize Clear and Simple Design:
     - Encourage students to keep their designs simple, avoiding clutter. Ask them to think about their user and how the design will support their experience.
     - Ask guiding questions such as:

       "How easy is it to read and interact with the app?"

       "Are the important features immediately visible and accessible?"
5. **Guiding Design with a User Persona**
   - Use the User Persona for Inspiration:
     - Remind students of the user persona they created in the Explore phase. Have them keep the persona's needs and challenges in mind while designing the wireframe.
     - Suggest they incorporate features that align with the persona's specific needs (e.g., larger buttons for users with motor impairments or clearer labels for those with cognitive challenges).
6. **Saving and Sharing Wireframes**
   - Save and Share the Work:
     - Show students how to save their work on Excalidraw and generate a link to share or submit their wireframe.
     - Encourage them to save and organize their wireframes as they will use these for reference in the Builder phase.
7. **Reflect on the Process**
   - Class Reflection or Exit Ticket:
     - At the end of the activity, have students reflect on their experience wireframing. Ask questions like:

       "What was the most challenging part of designing the wireframe?"

       "How did your wireframe evolve as you considered your user persona?"

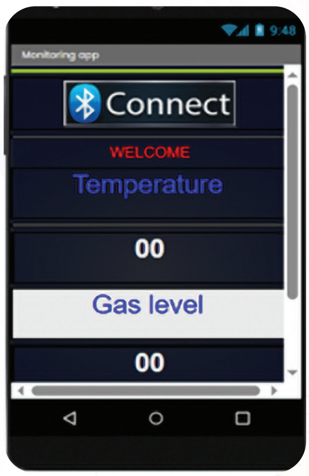
Builder

Objective:

Students will design the user interface (UI) of a Bluetooth-connected safety alert app in MIT App Inventor. They will create the layout for the app, including components for Bluetooth connection, sensor data display (temperature and gas levels), and alert notifications, focusing on clear, user-friendly design for real-time monitoring.

Activity Instructions:

1. **Open MIT App Inventor**
   - Go to MIT App Inventor and log in with your Google account.
   - Start a new project and name it SafetyAlertApp (or a relevant name).
2. **Set Up Your Screen Layout**
   - In the Designer tab, rename Screen1 to MainScreen (optional but helps with organization).
   - Set Align Horizontal to Center and Align Vertical to Top for a clean, centered layout.
3. **Add the Bluetooth Component**
   - BluetoothClient Component:
     - o From the Connectivity drawer, drag and drop a BluetoothClient component onto the screen.
     - o Rename it to BluetoothClient1.
   - Connect Button:
     - o Drag a Button component onto the screen, labeling it "Connect to Sensor" or something appropriate for Bluetooth connection.
     - o Rename the button to ConnectButton.
4. **Add Sensor Data Display Components**
   - Temperature Display:
     - o Drag a Label component and set its Text property to "Current Temperature: 0°F".
     - o Rename it to TempLabel.
   - Gas Level Display:
     - o Drag another Label component and set its Text property to "Gas Level: Safe".
     - o Rename it to GasLabel.
5. **Add Alert System**
   - Alert Notification (using Notifier):
     - o From the User Interface drawer, drag a Notifier component onto the screen. This will allow the app to show alerts when temperature or gas readings are unsafe.



- Rename it to AlertNotifier.

6. **Organize the Layout**
   - Use VerticalArrangement or HorizontalArrangement to neatly arrange your components:
     - Place the Temperature Display, Gas Level Display, and Connect Button in a VerticalArrangement to stack them vertically.
     - Make sure to leave enough space between components for a clean design.

7. **Adjust Component Properties**
   - Buttons and Textboxes:

     Adjust the width and height properties for better visibility. Ensure buttons are large enough for easy interaction, especially if the app is designed for elderly users.
   - Text Size:

     Increase the font size for Labels to ensure that temperature and gas levels are easily readable.

8. **Test the UI Design**
   - Save your work and test the layout using the MIT AI2 Companion or the emulator. Check for:
     - Proper alignment of components.
     - Readability of text.
     - Easy navigation of components (i.e., buttons and labels).

Possible Strategies for Implementation:

1. **Design with User Experience in Mind**
   - Simple and Clear Layout:
     - Focus on clarity and simplicity. The design should have large buttons for easy clicking, large text for readability, and visually distinct sections for each part of the app (temperature, gas levels, alerts).
   - Use Visual Cues:
     - Use color coding for safety: For instance, green for safe gas levels, red for dangerous gas levels, and blue for the temperature display.
     - Ensure contrast is strong enough for users with visual impairments.

2. **Prioritize Accessibility**
   - Size and Spacing:
     - Ensure that interactive components like buttons are large enough for easy tapping, especially if the app is intended for elderly users.
   - Clear Labels:
     - Label every component clearly and concisely to ensure users understand what each part of the app does. For example, label buttons like “Set Reminder”, “Connect to

Device", and "Check Sensor".

3. **Iterative Testing and Feedback**
   - Prototype Testing:
     - o Encourage students to test their UI design by clicking through the app and ensuring the layout is intuitive. They should ask themselves:

       "Is it easy to connect to Bluetooth?"

       "Are the sensor readings clear and easy to understand?"

       "Are alerts noticeable and actionable?"
   - Peer Feedback:
     - o Have students present their designs to a peer and ask for feedback on usability. One student could act as a "user," testing the app's navigation and interaction flow.
4. **Use Group Collaboration**
   - Collaborative UI Design:
     - o Divide the class into small groups, where each group works on one aspect of the app's interface (e.g., one group focuses on the Bluetooth button, another group focuses on the sensor display). This will help them engage with the design process while learning from others' approaches.
5. **Allow for Refinement**
   - Encourage Iteration:
     - o Remind students that design is an iterative process. After testing, they should refine the UI design based on usability feedback.
     - o Ask students to consider any difficulties they or their peers faced when navigating their design and adjust accordingly.

Mastermind

Objective:

Students will explore various wireless components beyond Bluetooth that could be integrated into their safety alert app. They will evaluate the advantages and disadvantages of each component and decide which might best suit their app's requirements, considering factors like range, power consumption, and ease of integration.

Answers:

Question: Can you think of other wireless components we could use in the app? What are they, and what are the advantages and disadvantages of each?

Possible Answers:

1. **Wi-Fi (Wireless Fidelity)**
   - Advantages:
     - o Longer Range compared to Bluetooth (typically up to 100 meters or more).

- o High Data Transfer Rate, making it suitable for larger data packets (e.g., images or large sensor data).
- o Reliable Connectivity when used in areas with strong Wi-Fi networks.

- Disadvantages:
  - o High Power Consumption, which could drain battery life quickly, especially on mobile devices.
  - o Requires access to a Wi-Fi network (which may not always be available in remote or outdoor areas).

2. NFC (Near Field Communication)

- Advantages:
  - o Low Power Consumption, ideal for battery-powered devices.
  - o Simple Pairing process (just need to bring the devices close together).
  - o Secure Communication, making it great for devices like medical or safety applications.
- Disadvantages:
  - o Short Range (typically less than 10 cm), limiting its use to very close proximity.
  - o Low Data Transfer Rate, making it unsuitable for sending large amounts of data.

3. Zigbee (Low Power, Short Range Wireless)

- Advantages:
  - o Low Power Consumption, making it suitable for battery-operated devices.
  - o Reliable Communication in crowded environments or IoT networks.
  - o Can support multiple devices in a network (mesh network).
- Disadvantages:
  - o Limited Range (about 10-100 meters, depending on conditions).
  - o Slower Data Transfer Rate compared to Wi-Fi or Bluetooth.

4. LoRa (Long Range)

- Advantages:
  - o Very Long Range (up to 15 km in open areas).
  - o Low Power Consumption, ideal for outdoor or remote applications.
  - o Can cover large geographical areas, making it great for IoT networks.
- Disadvantages:
  - o Slow Data Transfer Rate, making it less suitable for high-bandwidth applications like video or real-time data-heavy applications.
  - o Requires specific LoRaWAN infrastructure (gateways, base stations).

5. Infrared (IR) Communication

- Advantages:

- o Simple and Low-Cost solution for short-range communication.
- o Often used in remote control devices and can easily communicate with sensors or LEDs.

- Disadvantages:
  - o Very Short Range (typically a few meters).
  - o Line-of-Sight requirement, which limits its flexibility in some applications.
  - o Slower Data Transfer Rate.

6. **Cellular Network (4G/5G)**

- Advantages:
  - o Wide Range—can communicate over long distances anywhere with cellular coverage.
  - o Can handle large data transfers (e.g., streaming data or large alerts).
  - o Enables real-time communication and location tracking via GPS.
- Disadvantages:
  - o Higher Power Consumption compared to other wireless technologies.
  - o May incur costs associated with data transmission (e.g., mobile data plans).

Possible Strategies for Implementation:

1. **Research and Group Work**

- **Group Research:** Divide the class into groups, assigning each group a wireless component (Wi-Fi, NFC, Zigbee, LoRa, etc.) to research. Have each group present their findings, including the advantages, disadvantages, and ideal use cases.
- **Comparison Chart:** Create a visual chart comparing the different wireless components (range, power consumption, data rate, cost, etc.) to help students understand the trade-offs involved.

2. **Scenario-Based Decision Making**

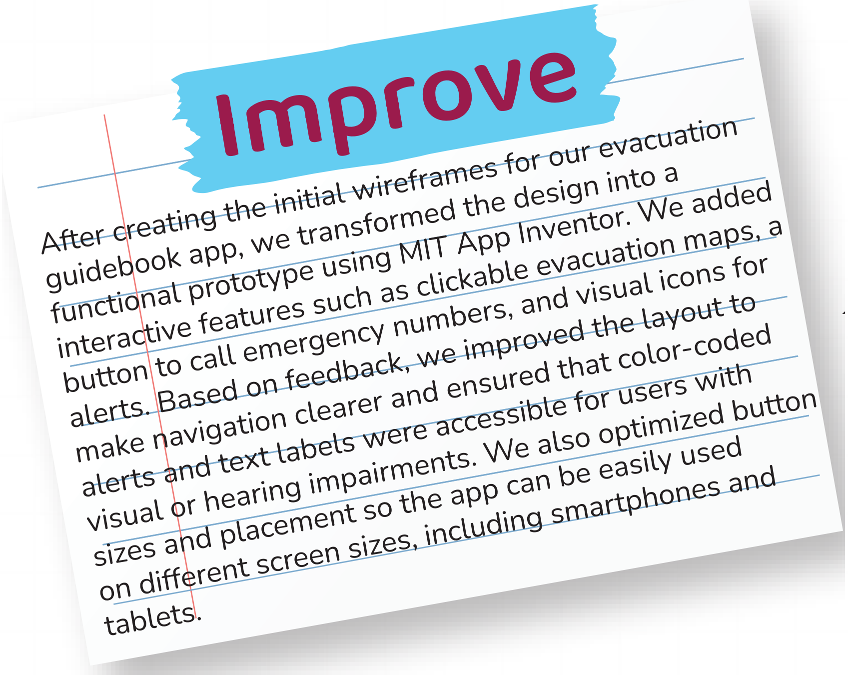
- **Use Case Scenarios:** Present students with a real-world scenario where they must choose the best wireless component for a particular application (e.g., a remote gas monitoring system, a wearable safety alert system, etc.).
- **Critical Thinking:** Encourage students to think critically about how each wireless technology would impact the app's design and functionality. For example:
  - o "If the app must operate in an area without Wi-Fi, would Bluetooth or LoRa be a better choice?"
  - o "For real-time temperature and gas readings with a large data transfer requirement, would Wi-Fi or Zigbee be more suitable?"

3. **Hands-on Prototyping**

- **Component Testing:** If feasible, provide students with small hardware kits (e.g., Bluetooth sensors, NFC tags, Zigbee modules, or LoRa boards) and let them experiment with each wireless technology. Have them build basic communication circuits to test how data is transferred and compare the ease of implementation for each technology.

4. **Evaluation Rubric**
   - **Design Evaluation:** Create a rubric for students to evaluate which wireless component is best for a safety alert system based on criteria such as range, power consumption, data transfer rate, and cost. Students can then use this evaluation to decide on the best component to incorporate into their app design.
5. **Application to App Design**
   - **App Integration:** Once students decide on the best wireless component, help them think through how it could be integrated into the app. For example:
     - o How will the app display data from the sensor connected via Bluetooth or Zigbee?
     - o How will the app notify the user when a dangerous gas level is detected or a high temperature is recorded using the chosen wireless technology?
   - Guide students to plan the UI/UX to accommodate the wireless features and ensure the app remains user-friendly.

Don't forget to go back to EDP paper and add it to improve part.



Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. **Introduce the Reflection**
   - Say: "Let's look back on everything we've explored this lesson. You learned how it

feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. **Guide Students Through the Checklist**
   - Read each statement aloud one at a time.
   - After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
   - Example prompts:
     - "Did you learn how people share ideas without talking?"
     - "Do you know how tools can help people talk?"
3. **Student Completion**
   - Let students check off or color the "Now I Can" boxes they feel good about.
   - Remind them: "It's okay if you didn't check all of them. We're all still learning."
4. **Optional Extension**
   - Ask students to circle the one they're most proud of.
   - Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. United Nations Sustainable Development Goals (SDGs)
   - **SDG 9:** Industry, Innovation, and Infrastructure

Students explore how wireless communication (Bluetooth) and hardware integration (Arduino) can enhance innovation, creating applications that contribute to smarter, safer environments such as home automation or environmental monitoring.

- **SDG 4:** Quality Education

Through hands-on learning with Bluetooth and Arduino, students develop technical and problem-solving skills, promoting inclusive, accessible education in STEM fields. This aligns with fostering critical thinking and innovation in technology-based education.

2. ISTE Standards for Students (International Society for Technology in Education)
   - **1. Empowered Learner**

Students design and build their own Bluetooth-enabled applications, taking ownership of their learning by solving real-world challenges (e.g., creating an alert system with Bluetooth communication).

- **4. Innovative Designer**

Students use MIT App Inventor and Arduino to create and test Bluetooth applications, learning to design solutions through iterative processes and refine their creations.

- **5. Computational Thinker**

By wiring Bluetooth modules to Arduino and developing mobile applications, students engage in logical sequencing and data transfer protocols, applying their computational thinking skills to real-world scenarios.

- **6. Creative Communicator**

Students develop a user-friendly app interface in MIT App Inventor, demonstrating how digital tools can communicate important information (e.g., alert systems based on sensor data).

3. NGSS – Engineering Design (Next Generation Science Standards)

- **MS-ETS1-1:** Define a problem and identify design criteria for a successful solution

Students define a problem related to wireless communication and Bluetooth connectivity, establishing design criteria for the app and hardware system (e.g., sensor data, alerts).

- **MS-ETS1-2:** Evaluate competing design solutions

Students evaluate different Bluetooth modules (e.g., HC-05 vs. HC-06) and app interface designs, considering factors like reliability, range, and ease of use for their app.

- **MS-ETS1-3:** Build and test a prototype

Students build a working Bluetooth system using Arduino, connecting sensors, establishing communication, and testing the app to ensure proper functionality and data transfer.

4. Computer Science & Digital Literacy Standards

(Aligned with CSTA and international ICT frameworks)

- **Algorithms & Programming**

Students write code to wirelessly transfer data between the Arduino and a mobile app, ensuring the Bluetooth module functions correctly for reading and transmitting sensor data.

- **Computing Systems**

Students understand the Bluetooth communication process, from wiring the Bluetooth module to connecting it to a mobile app, ensuring the system operates within its limits (range, connectivity).

- **Human-Computer Interaction**

Students design a user interface that is intuitive and easy to navigate, ensuring that the app's visual layout is accessible for the intended user.

- **Testing & Debugging**

Students test the Bluetooth connection, troubleshoot any issues with pairing or data transfer, and refine their design based on user feedback and test results.

5. 21st Century Skills (P21 Framework)

   - Critical Thinking

Students engage in problem-solving by diagnosing and correcting issues related to Bluetooth connectivity, sensor readings, and app functionality.

   - Collaboration

Students collaborate in teams to wire up Bluetooth modules, develop the app, and troubleshoot issues, working together to refine and improve the overall design.

   - Communication

Students explain the logic behind their designs and communicate their understanding of

Bluetooth and sensor integration through presentations and peer discussions.

- Creativity

Students personalize their app interfaces, designing unique solutions for different use cases while maintaining functionality and ease of use.